

Lec 10:

Recap:

① Random variables $X: \Omega \rightarrow \mathbb{R}$.

Well defined $X^{-1}((-\infty, x]) \in \mathcal{F}, \forall x \in \mathbb{R}$.

" $\{ \omega \in \Omega : X(\omega) \leq x \}$

$$\Rightarrow X^{-1}(\{x\}) \in \mathcal{F}$$

$$\Rightarrow X^{-1}((x_1, x_2)) \in \mathcal{F}$$

$$\Rightarrow X^{-1}([x_1, x_2]) \in \mathcal{F}$$

② Defined Probability Mass Function (PMF) $P_X(x) = P(X=x)$

$$X \sim \text{Bernoulli}(p) \rightarrow 1 - P_X(0) = P_X(1) = p$$

$$X \sim \text{Binomial}(n, p)$$

$$P_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

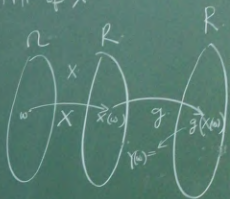
$$= P(\{\omega : X(\omega) = x\})$$

$$k \in \{0, 1, \dots, n\}$$

Function of a random variable $Y = g(X)$.

We want to find PMF of Y using PMF of X

$$P_Y(y) = \sum_{x \in \mathcal{X} | g(x) = y} P_X(x)$$



$$\begin{aligned} P_Y(y) &= P(Y=y) \\ &= P\left(\left\{ \omega \in \Omega \mid Y(\omega) = y \right\}\right) \\ &= P\left(\left\{ \omega \in \Omega \mid g(X(\omega)) = y \right\}\right) \end{aligned}$$

Let $\omega \in \Omega$
then $\{ \omega \in \Omega \mid Y(\omega) = y \}$

$P(\{ \omega \in \Omega \mid Y(\omega) = y \})$



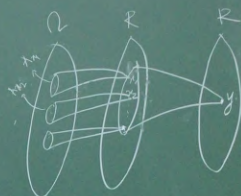
Example

$$\text{Let } A_x = \{\omega \in \Omega \mid X(\omega) = x\}$$

then

$$\{\omega \in \Omega \mid g(X(\omega)) = y\}$$

$$= \bigcup_{x \in X_0, g(x)=y} A_x$$



Clearly A_{x_1}, A_{x_2} are disjoint
for $x_1 \neq x_2$

$$P(\{\omega \in \Omega \mid g(X(\omega)) = y\}) = P\left(\bigcup_{x \in X_0, g(x)=y} A_x\right)$$

$$\sum_{x \in X_0, g(x)=y} P(X=x) \quad \Longleftrightarrow \quad \sum_{x \in X_0, g(x)=y} P(A_x)$$

Example: X be a RV taking values

$$X = \{-4, -3, \dots, 0, 1, 4\} \mid |X| = 9$$

$$P_X(x) = 1/9$$

$$Y = |X|, \quad \mathcal{Y} = \{0, 1, 2, 3, 4\}$$

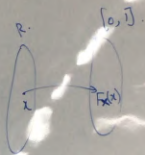
$$P_Y(0) = P_X(0) = 1/9$$

$$P_Y(1) = P_X(-1) + P_X(1) = 2/9$$

$$P_Y(y) = P_X(-y) + P_X(y) = 2/9 \quad y \in \{1, 2, 3, 4\}$$

Cumulative distribution function. (CDF) of a random variable X .

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= P(\underbrace{\{\omega \in \Omega \mid X(\omega) \leq x\}}_{X^{-1}((-\infty, x])}) \end{aligned}$$



Example:

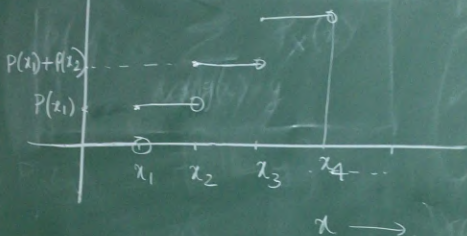
(a) X is a discrete random variable.

$$\begin{aligned} \mathcal{X} &= \{x_1, x_2, x_3, \dots\} \\ x_1 &< x_2 < x_3 < \dots \end{aligned}$$

Suppose $x_i \leq x < x_{i+1} \quad i \geq 1$.

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= \sum_{j=1}^i P(X = x_j) \end{aligned}$$

$F_X(x)$



$$P_X(x_i) = F_X(x_i) - F_X(x_{i-1})$$

$$\textcircled{2} \quad \Omega = [-1, 1] \quad P(a, b) = \frac{b-a}{2}$$

$$X(\omega) = \omega^2, \quad X \text{ takes values in } [0, 1]$$

$$\Rightarrow \mathcal{X}_0 = [0, 1]$$

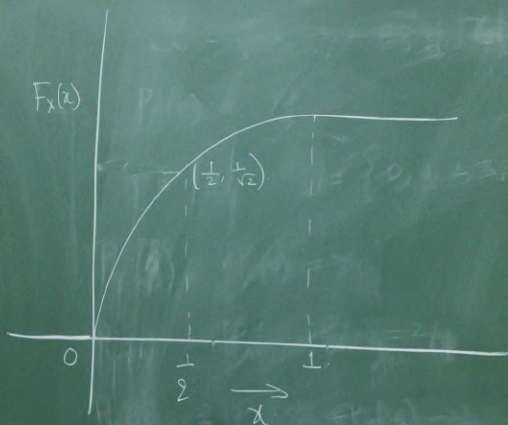
$$F_X(x) = \begin{cases} 0 & x < 0 \\ 1 & x > 1 \\ \sqrt{x} & x \in [0, 1] \end{cases}$$

For $x \in [0, 1]$

$$P(X \leq x) = P(\{\omega \in [-1, 1] \mid X(\omega) \leq x\})$$

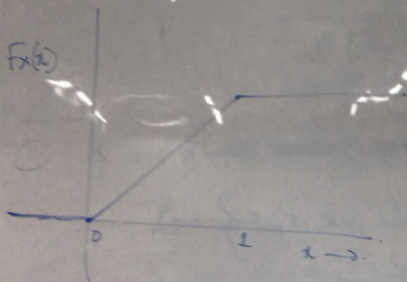
$$= P(\{\omega \in [-1, 1] \mid \omega^2 \leq x\})$$

$$= P(\{\omega \in [-\sqrt{x}, \sqrt{x}]\}) = (\sqrt{x} - (-\sqrt{x}))/2$$



$$(3a) \quad \Omega = [0, 1] \quad P((a, b)) = b - a$$

$$X(\omega) = \omega \quad \omega \in [0, 1]$$



$$F_X(x) = \begin{cases} 0 & x < 0 \\ x & x \in [0, 1] \\ 1 & x > 1 \end{cases}$$

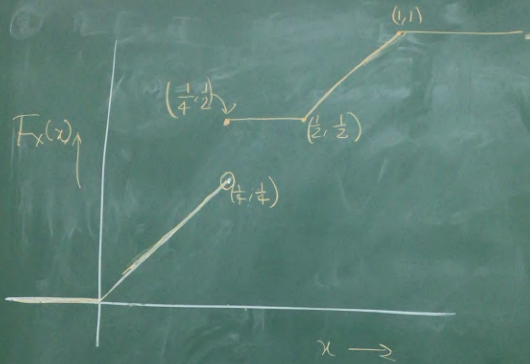
$$x \in [0, 1] \quad P(X \leq x) = P(\{\omega \leq x\}) = P(\omega \in [0, x])$$

$$(36) \quad X(\omega) = \begin{cases} \omega & \omega \in [0, \frac{1}{4}] \cup [\frac{1}{2}, 1] \\ \frac{1}{4} & \omega \in (\frac{1}{4}, \frac{1}{2}) \end{cases}$$

$$\begin{aligned} P(X \leq x) &= P(\{\omega \leq x\}) \quad x \in [0, \frac{1}{4}] \\ &= P([0, x]) \\ &= x \end{aligned}$$

$$\begin{aligned} P(X \leq \frac{1}{4}) &= P(\{\omega \in [0, \frac{1}{4}]\}) \\ x \in [\frac{1}{4}, \frac{1}{2}] &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} P(X \leq x) &= P(\{\omega \in [0, x]\}) = x \\ x \in [\frac{1}{4}, \frac{1}{2}] \quad P(X \leq x) &= \frac{1}{2} \end{aligned}$$



$$F_X\left(\frac{1}{4}\right) = \frac{1}{4}$$

$$F_X\left(\frac{1}{4}\right) = \frac{1}{2}$$

1/2

Properties of a CDF function

- ① If $x < y$, $F_X(x) \leq F_X(y)$ (Monotonic non-decreasing property)
- ② $\lim_{x \rightarrow -\infty} F_X(x) = 0$, $\lim_{x \rightarrow \infty} F_X(x) = 1$
- ③ $F_X(x) = F_X(x^+) \triangleq \lim_{h \rightarrow 0^+} F_X(x + \frac{1}{h})$ (Right Continuity)
- ④ $P(X > x) = 1 - F_X(x)$
- ⑤ $P(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$
- ⑥ $P(X = x) = F_X(x) - F_X(x^-)$

Where $F_X(x^-) = P(X < x)$

$$(1) F_X(x) = P(\overbrace{\{\omega \in \Omega : X(\omega) \leq x\}}^{= A_x}).$$

$$F_X(y) = P(\underbrace{\{\omega \in \Omega : X(\omega) \leq y\}}_{= A_y}).$$

$$A_x \subseteq A_y \quad \text{as } \omega \in A_x \Rightarrow X(\omega) \leq x$$

but we know $x \leq y$

$$\Rightarrow X(\omega) \leq y \Rightarrow \omega \in A_y.$$

$$\Rightarrow A_x \subseteq A_y.$$

$$F_X(x) = P(A_x) \leq P(A_y) = F_X(y).$$

$$(4) \quad \begin{aligned} P(X > x) &= P(\{\omega : X(\omega) > x\}) \\ &= P(A_x^c) \\ &= 1 - P(A_x) = 1 - F_X(x). \end{aligned}$$

$$\textcircled{5} \quad P(x_1 < X \leq x_2) = P(A \cap B^c)$$

$$A = \{\omega \in \Omega \mid X(\omega) \leq x_2\} = P(A) - P(A \cap B)$$

$$B = \{\omega \in \Omega \mid X(\omega) \leq x_1\} = P(A) - P(B)$$

$$= F_X(x_2) - F_X(x_1)$$

$$B \subseteq A \Rightarrow A \cap B = B$$

$$\textcircled{6} \quad P(X=x) = P(A \cap B^c)$$

$$A = \{\omega \in \Omega \mid X(\omega) \leq x\} = P(A) - P(B)$$

$$= F_X(x) - F_X(x)$$

$$B = \{\omega \in \Omega \mid X(\omega) \leq x\}$$

Continuity properties

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P\left(\bigcup_{i=1}^n A_i\right).$$

Corollary 1: $A_1 \subseteq A_2 \subseteq A_3 \dots$

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n).$$

Corollary 2: $A_1 \supseteq A_2 \supseteq A_3 \dots$

$$P\left(\bigcap_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n).$$

$$(2) \lim_{x \rightarrow -\infty} F_X(x) = 0$$

$$\begin{aligned} \lim_{x \rightarrow -\infty} F_X(x) &= \lim_{n \rightarrow \infty} F_X(-n) = \lim_{n \rightarrow \infty} P(X \leq -n) \\ &= \lim_{n \rightarrow \infty} P\left(X^{-1}((-\infty, -n])\right) \end{aligned}$$

$$A_n = X^{-1}((-\infty, -n])$$

$$A_1 \supseteq A_2 \supseteq A_3 \dots \quad \bigcap_{n=1}^{\infty} A_n = \emptyset$$

$$B_n^c = (-\infty, -n]$$

$$\text{let } x \in \mathbb{R}, \exists n \in \mathbb{N} \text{ st } x > -n \Rightarrow x \notin B_n$$

$$\bigcap_{n=1}^{\infty} B_n = \emptyset \Rightarrow \bigcap_{n=1}^{\infty} A_n = \emptyset$$

$$\lim_{x \rightarrow -\infty} F_X(x) = \lim_{n \rightarrow \infty} P(A_n)$$

$$A_1 \supseteq A_2 \supseteq \dots = P\left(\bigcap_{n=1}^{\infty} A_n\right) \text{ (from Corollary 2)}$$

$$= P(\emptyset) = 0$$

2b left an exercise.

$$\lim_{x \rightarrow \infty} F_X(x)$$

$$= \lim_{n \rightarrow \infty} P(A_n)$$

$$= P\left(\bigcup_{n=1}^{\infty} A_n\right) \text{ (by Corollary 3)}$$

$$= P(\mathbb{R}) = 1.$$

$$A_n = X^{-1}([-\infty, n])$$

$$A_1 \subseteq A_2 \subseteq \dots$$

$$\bigcup_{n=1}^{\infty} A_n = \mathbb{R}$$

$$1/10^2 \in$$

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